CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 8: Remainder and Factor Theorems

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Find the remainder when $x^3 + 2x^2 - 5x + 3$ is divided by $(x - 1)$.

2. Find the remainder when $2x^3 - x^2 + 3x - 4$ is divided by $(x + 2)$.

3. Find the remainder when $4x^3 - 4x^2 + x + 2$ is divided by $(2x - 1)$.

4. Show that $(x + 3)$ is a factor of $x^3 + 2x^2 - 5x - 6$.

5. Find k if $(x - 3)$ is a factor of $x^3 - kx^2 + 2x - 15$.

6. Find a if $x^3 + ax^2 - 2x + 7$ leaves remainder 13 when divided by $(x - 2)$.

7. When $x^3 + 3x^2 - mx + 4$ is divided by $(x - 2)$, the remainder is $m + 3$. Find m.

8. If $(x - 2)$ and $(x + 1)$ are both factors of $x^3 + ax^2 + bx - 2$, find a and b.

9. Factorise $x^3 - 6x^2 + 11x - 6$ completely.

10. Factorise $x^3 + 2x^2 - 5x - 6$ completely and hence solve $x^3 + 2x^2 - 5x - 6 = 0$.

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Answer Key

1. $1 + 2 - 5 + 3 = 1$
2. $-16 - 4 - 6 - 4 = -30$
3. $\frac{1}{2} - 1 + \frac{1}{2} + 2 = 2$
4. $f(-3) = -27 + 18 + 15 - 6 = 0$
5. $27 - 9k + 6 - 15 = 0$, so $k = 2$
6. $8 + 4a - 4 + 7 = 13$, so $a = \frac{1}{2}$
7. $24 - 2m = m + 3$, so $m = 7$
8. $4a + 2b = -6$ and $a - b = 3$ give $a = 0, b = -3$
9. $(x - 1)(x - 2)(x - 3)$
10. $(x + 1)(x - 2)(x + 3)$; $x = -1, 2, -3$

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